Sun versus Clock: Formalizing the Hindustani Samay System on Solar Phase and Measuring Its Divergence from the Fixed-Clock Convention

Sudipta Bhattacharya
IEM Centre of Excellence for Cloud Computing and IoT,
Department of CSE (AIML) and CSBS, Institute of Engineering and Management,
Kolkata, West Bengal, India
bhattacharya.sudipta@gmail.com

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Abstract

In Hindustani classical music, ragas carry prescribed performance windows (*samay*). The genuinely solar part of this prescription is the *sandhi-prakash* (twilight) class, anchored to sunrise and sunset, which shift with latitude, longitude, and season, while prior computational treatments of raga time assume a fixed clock. We propose a solar formalization of the samay system: six window arcs defined on an astronomically computed, season-invariant solar phase $\psi(t, \ell, d)$, with the twilight arcs straddling the solar events, presented explicitly as a modelling convention consistent with one reading of prahar reckoning. By exact integration of the disagreement set, the fixed-clock convention and the solar formalization assign different windows for 15.7% of the day weighted by state population (15.9% under a 1-km population raster), rising to 35.9% at the western extreme in winter, with window onsets shifting ≈81 minutes between Kolkata and Srinagar under one time zone and ≈80 minutes across seasons at a single site. We release a unified, source-cited raga–samay table (91 ragas with explicit anytime, seasonal, and imported classes, reconciling disagreements in prior engineering sources). On Saraga Hindustani, features derived from the corpus's expert melody and tonic annotations (tonic-normalized cent-resolution pitch distributions) identify ragas at $89.3\% \pm 5.1$ recording-level majority vote over 8 classes under leave-one-artist-out evaluation, above recording-level permutation chance at $p \leq 10^{-4}$ on every seed, and 13.1 points above a 12-bin pitch-class baseline; whether Bhatkhande's codified svara–samay correlation is audible remains preliminary, because in this corpus most time-windows are carried by a single raga, collapsing the window task into raga identification. A closed-form identifiability analysis of the formalization itself is confined to an appendix. We state plainly what the framework does not yet show: recommendation value, performance-practice claims, and affective claims await timestamped performance data and a psychometric rating study, both specified herein.

Keywords. Indian classical music; raga; samay; computational musicology; solar time; music information retrieval; cultural analytics.

1 Introduction

Hindustani classical music assigns each raga a prescribed window in the diurnal cycle. A listener at a morning concert expects Todi or Jaunpuri; Yaman belongs to the first hours of night; Lalit announces the end of night. This *samay* (time) system, systematized in the early twentieth century by V. N. Bhatkhande, has few parallels in other art-music traditions, and it has begun to attract computational attention: Roy, Chakrabarty and De [1] trained a neural network to

classify ragas into day and night classes from symbolic features, using fixed clock windows such as 07:00–10:00.

This paper starts from an observation about that fixed clock. The part of the samay system with the clearest physical referent is the *sandhi-prakash* class: the twilight ragas, defined by the transition of light at dawn and dusk. Sunrise and sunset are functions of latitude, longitude, and date. Across India, which spans roughly 29° of longitude under a single time zone, sunrise at the same IST clock instant differs by nearly two hours between the eastern and western extremes; at one site, it shifts by some 80 minutes between the solstices. A computational model that encodes samay as fixed clock intervals therefore cannot be simultaneously faithful to the tradition's twilight anchoring in Srinagar and in Kolkata, in June and in December. The observation licenses less than it may seem to. The tradition itself does not legislate per-location astronomical recomputation: Bhatkhande states prahars and approximate clock times, and All India Radio historically scheduled by clock. We therefore propose solar anchoring as an explicit formalization, strongest for the sandhi-prakash windows, and we quantify its divergence from the fixed-clock convention rather than declaring either one correct.

Concretely, we make four contributions. **(i)** A solar formalization of samay: a season-invariant solar phase $\psi(t, \ell, d)$ computed from standard solar-position equations, on which the six traditional performance windows are defined as arcs, the twilight arcs straddling sunrise and sunset (Section 3). By exact integration, this formalization disagrees with the fixed-clock convention for 15.7% of the day weighted by state population, 18.8% on average across 13 cities spanning the extremes, and up to 35.9% in the far west in winter. **(ii)** A unified, source-cited raga–samay table of 91 ragas with explicit *anytime*, *light*, *seasonal*, and *imported* classes, reconciling errors and inconsistencies we document in prior engineering sources (Section 4). **(iii)** A study on Saraga Hindustani [12] with annotation-derived features under leave-one-artist-out evaluation. Tonic-normalized cent-resolution pitch distributions identify ragas at $89.3\% \pm 5.1$ recording-level majority vote over 8 classes, with recording-level permutation $p \leq 10^{-4}$ on every seed, and the companion question, whether the codified svara–samay correlation is audible, proves underdetermined in this corpus; we report it as preliminary and state its confound (Section 5). **(iv)** A closed-form identifiability analysis establishing that the formalization is recoverable from context features, with the Bayes-optimal rule as the ceiling; because its labels are generated by the formalization itself, it is a design check, not evidence, and it is confined to Appendix A.

Throughout, we treat the samay prescriptions as the tradition's normative claims, not as established facts about listeners, and we treat samay in any deployed system as a soft prior over a ranked list rather than a hard filter: modern concert practice freely programs morning ragas at night, and Bhairavi concludes concerts at any hour. The falsifiable version of this paper's central premise, that solar-corrected samay predicts actually-programmed ragas better than the fixed clock, requires timestamped performance data; Section 7 sets it out as the main direction for future work.

2 Background and related work

The samay system and its scholarship. The association of ragas with times of day is attested from the medieval gāna-kāla rules of the *Saṅgīta-Ratnākara* lineage, but the system in modern use is substantially Bhatkhande's rationalization [2]: ragas are grouped by the position of the vādī (sonant) in pūrvāṅga or uttarāṅga, by the flattened second and sixth degrees that mark the sandhi-prakash (twilight) classes, and by the madhyam as an indicator distinguishing day from night ragas. The two systems differ materially and cannot be cited interchangeably. For our purposes the important point is that the scholarship on the system (Jairazbhoy's analysis of its internal logic [3], Deva [4], Kaufmann's per-raga compendium [5], Bor et al.'s performance-time listings [6], and Clayton on time in Indian music generally [7]) treats its rationale as conventional and contested, and documents a persistent gap between prescription and modern concert practice,

in which evening programming dominates and the concluding raga Bhairavi is sung at any hour. We therefore model samay as a normative prescription whose structure can be formalized and whose observance is an empirical question, not a ground truth.

Computational raga modelling. Raga recognition is a mature MIR task on CompMusic-lineage corpora: pitch-class distributions [8], fine-grained cent-resolution pitch distributions with kernel density estimation [9], time-delayed melody surfaces (TDMS) [10], which remain the standard strong baseline, and phrase- and deep-learning-based approaches. All model raga identity from tonic-normalized pitch representations; none models performance context such as time or place. The one computational treatment of raga time, Roy et al. [1], classifies 23 ragas into two classes (pūrvāṅga/uttarāṅga rendered as day/night) from three symbolic features under fixed clock windows; we document in Section 4 that several of its raga–window assignments do not survive comparison with the musicological sources.

Context-aware recommendation and music emotion. Context-aware recommender systems have long modelled time and location [13], including for music [14]; our claim is therefore scoped: location is unmodelled within computational raga systems, not in music recommendation at large. Music emotion recognition supplies mature machinery (categorical and dimensional valence-arousal labels, benchmark protocols, and the finding that such labels require controlled annotation [11]), and raga-specific emotion has an empirical literature of its own [15–17]. This paper makes no affective claims: a psychometric rating study is specified as future work, and the synthetic tokens used in the identifiability analysis of Appendix A are labelled as such wherever they appear.

The gap. Within computational raga modelling: (a) time has been modelled only as a fixed clock; (b) location has not been modelled at all, although the tradition's twilight anchor is a solar phenomenon that varies with location and season; (c) no unified, source-cited machine-readable raga–samay resource existed. Sections 3–5 address (a)–(c) in turn. **[Pre-submission: attach the documented search protocol (databases, queries, dates) as supplementary material backing claim (b).]**

3 A solar formalization of samay

Solar phase. For a location $\ell = (\varphi, \lambda)$ (degrees, east-positive), date with day-of-year N, and local clock time t on a zone with reference meridian λ_{tz} (IST: 82.5°E), we use the standard approximations for solar declination and the equation of time,

$$\delta(N) = 23.44^\circ \sin\Big(\tfrac{360^\circ}{365}(N-81)\Big), \qquad E(N) = 9.87\sin 2B - 7.53\cos B - 1.5\sin B, \quad B = \tfrac{360^\circ}{365}(N-81), \tag{1}$$

local solar time and the sunrise hour angle,

$$T_{\mathrm{sol}} = t + \frac{4(\lambda - \lambda_{tz}) + E(N)}{60}, \qquad \cos\omega_0 = -\tan\varphi\tan\delta, \qquad D = \frac{2\omega_0}{15^\circ}, \;\; \bar{D} = 24 - D, \tag{2}$$

with sunrise/sunset at $12 \mp \omega_0/15$ in solar time ($\cos\omega_0$ clipped to $[-1, 1]$; atmospheric refraction, about 0.83° or 3–4 minutes, is an optional term, off in all experiments). The prahar index and the season-invariant solar phase $\psi \in [0, 1)$, with day mapped to $[0, 0.5)$ from sunrise and night to $[0.5, 1)$ from sunset, follow as

$$\Pi = \begin{cases} \lceil 4(T_{\mathrm{sol}} - t_{sr})/D \rceil, & \text{day} \\ 4 + \lceil 4\Delta/\bar{D} \rceil, \;\; \Delta = (T_{\mathrm{sol}} - t_{ss}) \bmod 24, & \text{night} \end{cases} \qquad \psi = \begin{cases} \dfrac{T_{\mathrm{sol}} - t_{sr}}{2D}, & \text{day} \\ \dfrac{1}{2} + \dfrac{\Delta}{2\bar{D}}, & \text{night} \end{cases} \tag{3}$$

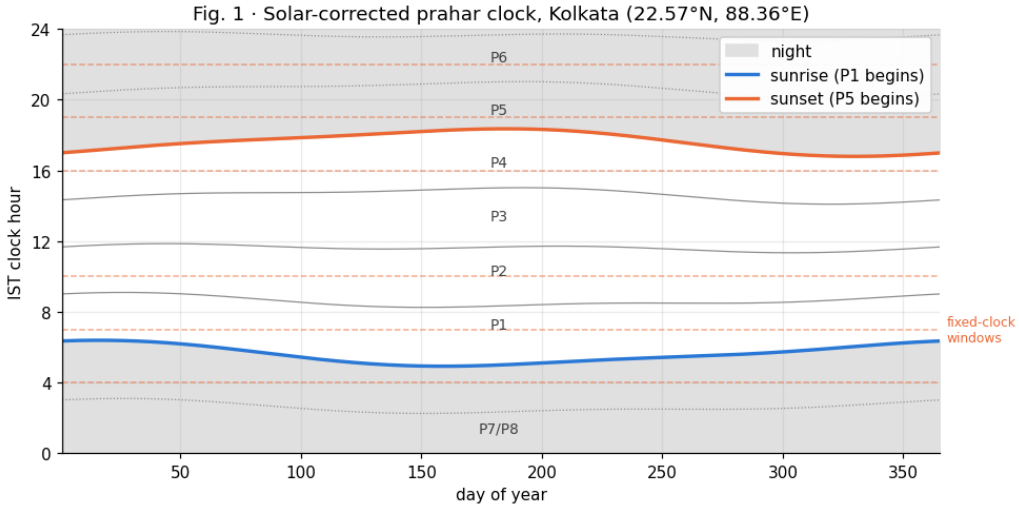


Figure 1: The solar window geometry over a year at Kolkata (22.57°N, 88.36°E). Solid curves: sunrise and sunset; the eight prahar boundaries breathe with the seasons, while the fixed-clock windows of prior work (dashed) stay frozen.

with model features $x_{\text{time}} = [\sin 2\pi\psi, \cos 2\pi\psi, \text{onehot}(\Pi)]$. Equation (3)'s quarter-of-daylight prahar reading is one convention among several (pañcāṅga reckonings differ); we adopt it as an explicit modelling choice, and Roy et al.'s two classes correspond to $\Pi \in \{1\text{–}4\}$ vs $\{5\text{–}8\}$ by construction, not as a validation. Against the full NOAA solar algorithm over a $6 \times 6 \times 12$ grid spanning India, these approximations err by 0.9 minutes on average for sunrise (maximum 3.7) and 0.7 minutes for sunset (maximum 2.5), negligible beside the window widths.

Window arcs. The six traditional windows are defined as arcs of ψ: dawn $[0.95, 0.04)$, morning $[0.04, 0.17)$, midday $[0.17, 0.42)$, dusk $[0.42, 0.55)$, evening $[0.55, 0.70)$, night $[0.70, 0.95)$. The two sandhi-prakash arcs straddle their solar events ($\psi = 0$ is sunrise, $\psi = 0.5$ sunset): Lalit, Sohini, Paraj and Bhatiyar belong to the end of night, and Marwa and Shree span sunset. A hard boundary at the solar instant would misplace precisely the ragas whose anchoring is genuinely solar. "Mid-morning" thus denotes the same phase in June and December, the invariance the prescription intends, while the raw clock does not. Figure 1 shows the resulting window geometry over a year at Kolkata against the fixed-clock windows of prior work, and Figure 2 the spread of the first prahar's onset across 13 cities under one IST clock.

Divergence between the two conventions. Neither the fixed-clock windows of prior work nor our solar arcs are legislated by the tradition, so we quantify their disagreement rather than declare error. The comparator is specified exactly: the fixed-clock convention is the six IST windows of Roy et al. [1] (dawn 04–07, morning 07–10, midday 10–16, dusk 16–19, evening 19–22, night 22–04), applied identically at every location and date; the solar convention is the arc set above, fixed from prahar structure before this comparison was run and not fitted to it. Because both conventions are piecewise constant in clock time, their disagreement over a day is a union of intervals delimited by at most twelve breakpoints (six clock boundaries and six arc boundaries mapped to clock time), so we integrate it exactly rather than sample it; an earlier grid-sampled figure was inflated by boundary-aligned sampling, and the exact values replace it. Across 13 Indian cities $\times$ 12 months, the conventions assign different windows for 18.8% of the day on average (minimum 7.0%, median 18.5%, maximum 35.9%, at Bhuj in February; Figure 3). Two analytic facts drive this: at the 21 December solstice, sunrise differs by 81 minutes between Kolkata and Srinagar under one IST clock, and by 80 minutes at Kolkata alone between solstices;

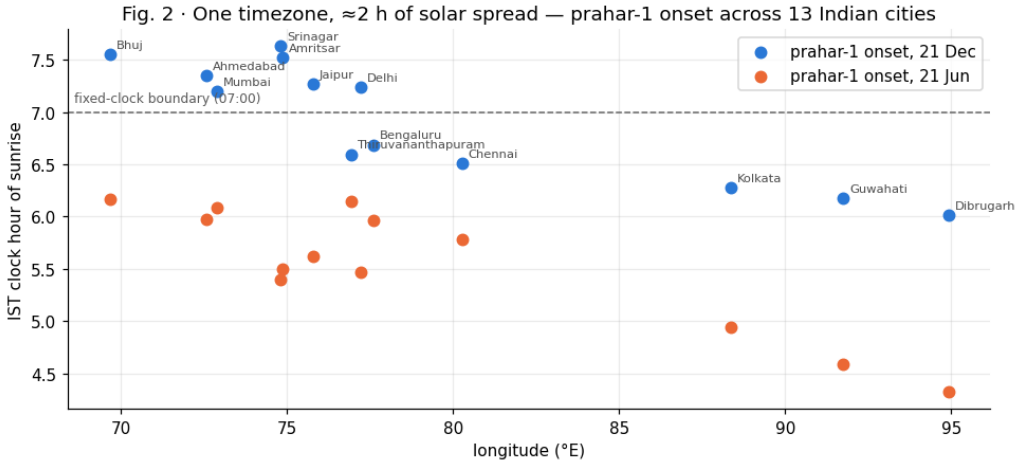


Figure 2: One time zone, about two hours of solar spread: the onset of the first prahar (sunrise) in IST clock time across 13 Indian cities at the two solstices, against the 07:00 fixed-clock boundary of prior work.

Table 1: Worked example of the solar geometry (no refraction; all clock times IST).

Site, date	Sunrise	Sunset	Daylight D
Kolkata, 21 Dec	06:17	16:54	10 h 37 m
Srinagar, 21 Dec	07:38	17:21	9 h 43 m
Kolkata, 21 Jun	04:56	18:20	13 h 24 m

the pure longitude span of India contributes about 115 minutes (Table 1). The 13-city sample deliberately includes the geographic extremes. Weighted by state population (Census 2011, 30 states represented by their capitals), the divergence is 15.7% of the day, and re-weighting by a 1-km population raster [20] (1,127 aggregated cells above 50,000 people, covering 1.38 billion) gives 15.9%, so the state-capital approximation is sound. Across states the figure ranges from 10.0% (Odisha) to 24.4% (Gujarat). The headline is also not an artifact of any single boundary choice: shifting each of the six arc boundaries by ±0.02 (about ±29 minutes of a 12-hour day) moves the 13-city mean within 17.3–20.7%. More than one hour in seven of the median Indian listener's day is assigned differently by the two conventions: whatever one's reading of samay, the choice is consequential, and the rest of the paper examines that premise.

4 A unified, source-cited raga–samay table

Computational work on raga time has so far inherited its raga–window assignments from engineering papers rather than musicological sources, and the inherited tables contain errors a knowledgeable reader will catch immediately: in [1], Gara (a light, thumri-aṅga raga of the evening) appears in the morning; Bhairavi, the sarva-kālika concluding raga performed at any hour, is assigned to midday; Gaud Sarang (conventionally early afternoon) to morning; Jaunpuri and Bilaskhani Todi (late morning) to midday. We therefore built a unified table of 91 ragas, released as machine-readable CSV, with per-raga sources (Bhatkhande's *Kramik Pustak Mālikā* [2]; Kaufmann [5]; Bor et al. [6]) and a dispute-notes column recording where the sources disagree.

Each raga carries a status: *fixed* (60 ragas; time-supervisable), *anytime* (Bhairavi), *light* (8: thumri-aṅga vehicles such as Pilu, Kafi, and Gara, flexible by practice), *seasonal* (10: the Malhar family, Basant and Bahar, the monsoon-associated Desh, and the spring Hindol, where

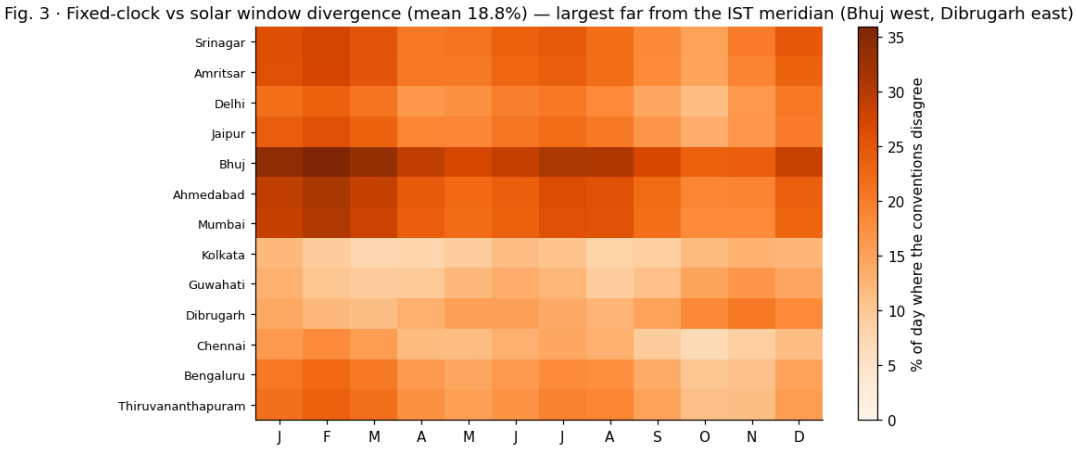


Figure 3: Fixed-clock vs solar window divergence (% of the day, exact integration) per city and month. The divergence is largest far from the 82.5°E IST meridian (Bhuj in the west, Dibrugarh in the east) and in winter.

the seasonal association dominates the diurnal one), and *imported* (6: Carnatic and modern ragas such as Hansadhwani, Kirwani, Abhogi, and Jog, whose "times" are modern concert habit rather than traditional samay). Only *fixed* ragas enter time-supervised experiments; all statuses enter raga identification. Exclusion criteria are thus applied uniformly, a consistency the earlier ad-hoc tables lacked. Where our window assignment required adjudication (Durga and Shankara sit on the evening/night discretization boundary; Maru Bihag, a raga distinct from Bihag despite the name, keeps its own entry in the night class; Sohini is placed on the pre-sunrise side of the dawn arc), the table records the decision and its source rather than hiding it. Saraga's IAST-transliterated spellings are resolved to table entries by diacritic-stripped matching plus an explicit alias list, and every merge is an auditable editorial decision (for example, Kalyāṇ to Yaman, and Todi to Miyan ki Todi). **[Pre-submission: page-level references from library copies; worksheet in progress.]**

5 Experiments on Saraga Hindustani with annotation-derived features

Data and features. Saraga 1.5 Hindustani [12] has 108 concert recordings, of which 93 carry raga metadata resolving to 60 distinct ragas of the unified table (4 rare names remain unmapped and are listed in the released code). Ragas with at least 3 recordings form the identification set: $N = 28$ recordings, $K = 8$ ragas (Shree and Bhairavi 5 each; Yaman, Jog, Lalit, Marwa, Miyan Malhar, and Miyan ki Todi 3 each), 8 distinct artists, 903 thirty-second segments, 7.5 hours. A raga is defined by its svara material relative to the tonic, so the features are tonic-normalized cent-resolution pitch distributions: 120-bin folded histograms (10-cent resolution) computed from Saraga's expert predominant-melody and ctonic annotations, plus a voicing ratio. We say annotation-derived rather than audio-based advisedly: no representation is learned from the waveform, and the results certify the pipeline from expert melody annotations to raga identity, not an end-to-end audio system. This is the representation family of the raga-recognition literature [8–10]. It is intonation-aware where 12-bin chroma is not, and it avoids the tabla and tanpura contamination of full-mix spectral features. A speech SSL baseline (wav2vec 2.0 [19], mean-pooled) scored at chance on this corpus in a preliminary run, the expected failure of a speech representation under small per-class counts; we report it as a cautionary comparison, not a contribution. A sanity check of the tonic annotations (histogram mass concentrated at Sa and the fifth) flags none of the 28 recordings, so we cite the ctonic files as verified.

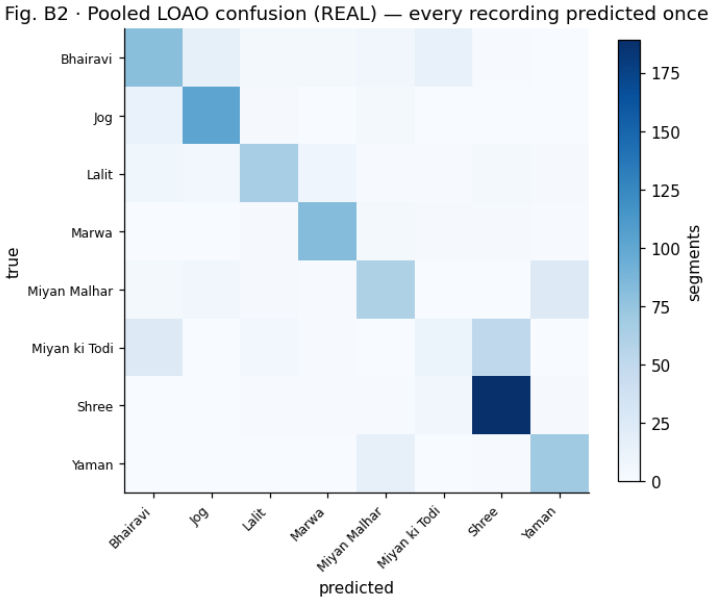


Figure 4: Pooled LOAO segment confusion over the 8-raga identification set; every recording is predicted by a model that never saw its artist. The dominant off-diagonal mass is the allied pair Miyan ki Todi → Shree.

Protocol. Evaluation is leave-one-artist-out (LOAO) over the 8 artists: in each fold, every recording by the held-out artist is predicted by a model that has never seen that performer, so performer voice and venue acoustics cannot leak; a validation carve-out within each fold is itself artist-disjoint. Pooling the folds predicts every recording exactly once. The primary metric is recording-level majority vote over segment predictions, since the recording, not the 30-second segment, is the musical unit; segment accuracy and class-balanced segment accuracy are secondary. Segments from one recording are statistically dependent, so segment-level permutation tests overstate significance; chance is instead estimated by permuting recording labels (10^4 shuffles, $p = (c+1)/(n+1)$). The whole protocol is repeated over three seeds, which vary initialization and the validation carve-out, and we report mean ± sd.

Raga identification. Recording-level majority vote $89.3\% \pm 5.1$ (per-seed 85.7, 96.4, 85.7), with recording-level permutation $p \leq 10^{-4}$ on every seed against an empirical chance of about 13.5%; segment accuracy $74.0\% \pm 5.9$, balanced $70.8\% \pm 6.6$. Figure 4 shows the pooled confusion. Eight classes and 28 recordings make this a small study, and its value lies in what it certifies: the field's standard representation, computed from expert annotations under the strictest split in use, recovers raga identity far above chance on an open corpus, and the same pipeline applies to larger corpora without modification. Two further analyses support the representation choice. Against the classic 12-bin pitch-class distribution of Chordia and Rae [8], computed from the same cached histograms under the same LOAO protocol ($76.2\% \pm 4.5$ recording-level), cent resolution adds 13.1 points; removing the voicing ratio changes nothing (+0.0), evidence that the feature is not smuggling in recording conditions. The error analysis behaves as distribution features predict: pooling segment errors over the three seeds (705 in all), 41% fall in scale-sharing allied pairs, led by Miyan ki Todi taken for Shree (128 errors; the two share komal re and tivra ma, and Miyan ki Todi's recall, 0.11 in the last seed, is the casualty), with Lalit–Marwa and Bhairavi–Miyan ki Todi following; the largest confusion outside the listed pairs, Miyan Malhar with Yaman in both directions, is not scale-based and plausibly reflects the seasonal raga's heterogeneous melodic material. Distribution features discard melodic order, so allied-raga confusion is the expected failure mode, and sequence-aware baselines such as TDMS [10] are the right next comparison.

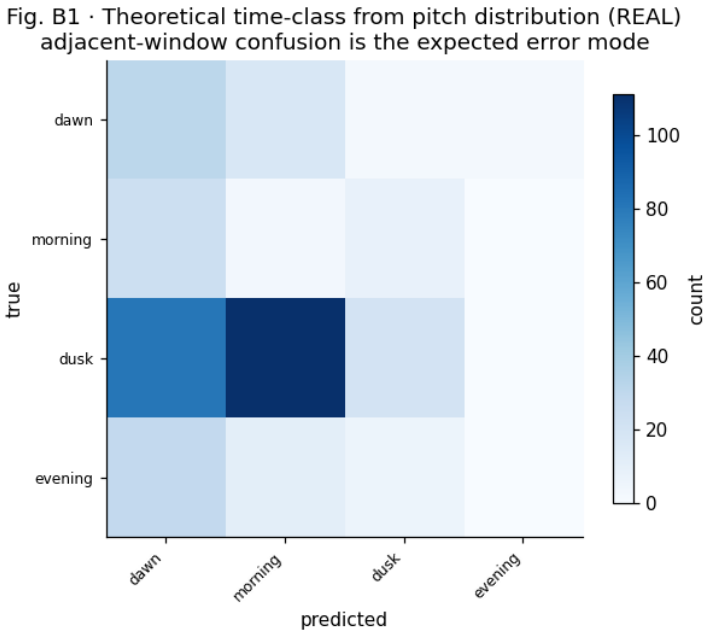


Figure 5: Theoretical time-class from pitch distributions on Saraga (preliminary; see text for the single-raga-window confound).

Is the svara–samay correlation audible? (preliminary). Bhatkhande's system correlates time-class with svara material by design: komal re and dha mark the sandhi-prakash classes, and the madhyam separates day from night ragas. One may therefore ask whether a classifier recovers a raga's theoretical time-class from pitch content alone. This is a question about the codification, not about when Saraga's concert recordings were performed. On the fixed-status subset (17 recordings, 5 ragas, 4 windows) the answer is underdetermined: 39.0% mean top-1 with per-seed permutation p of 10^{-4}, 0.002, and 0.92, and, decisively, three of the four windows are carried by a single raga (dawn by Lalit, morning by Miyan ki Todi, evening by Yaman), so the window task largely collapses into raga identification (Figure 5). We report this experiment as preliminary with the confound stated, and identify the remedy (a corpus with at least two ragas per window, Section 7) rather than claiming the result.

What we do not claim. No context-fusion result on real audio is reported. Saraga carries no performance timestamps, and fusing audio with context sampled from the raga's own canonical arc would constitute label leakage by construction. The released notebook contains such a cell solely as a pipeline check, labelled accordingly.

6 Discussion

Prescription and practice. Any computational treatment of samay must face the fact that the prescription is imperfectly observed. Concert life concentrates in the evening, broadcast schedules have historically run by clock, and the tradition itself sanctions exceptions: Bhairavi concludes at any hour, and seasonal ragas override the diurnal cycle in season. We have therefore been careful to let the system's structure and its observance come apart. The solar formalization (Section 3) models the structure; whether practice follows it is an open empirical question that our framework makes testable but that Saraga, lacking timestamps, cannot answer. For the same reason a deployed selector must treat samay as a soft prior over a ranked list. Our reference implementation returns probabilities, always admits the anytime class, and never refuses a raga,

since a system that forbade Todi at 8 p.m. would contradict a century of concert practice while claiming to honour tradition.

Limitations. (i) The identifiability analysis of Appendix A recovers our own construction; its value is sensitivity analysis, not validation. (ii) The Saraga study is small ($K = 8$, $N = 28$) and its time-class companion is confounded by single-raga windows; both scale with corpus size, not with method changes. (iii) Its features derive from expert melody annotations, so the result certifies the representation and protocol, not an end-to-end audio system; predominant-melody extraction would have to replace the annotations for deployment. (iv) Distribution features discard melodic grammar (āroha/avaroha, pakad, gamaka), so allied ragas sharing a scale set remain hard, and sequence-aware baselines such as TDMS [10] are the right next comparison. (v) The quarter-of-daylight prahar reading and the arc boundaries are conventions; the reported sensitivity band (17.3–20.7% for the 13-city mean under ± 0.02 boundary shifts) should accompany any claim built on them. (vi) The 13-city divergence sample is unweighted and spans the extremes by design; the population-weighted figures (15.7% by state capitals, 15.9% by raster) are the representative ones. (vii) No affective claims are made anywhere: the rasa tokens of Appendix A are synthetic, and the rating-study protocol (enculturated listeners, navarasa multi-label plus valence-arousal, inter-rater reliability thresholds) is specified separately for the follow-up study.

7 Future work and conclusion

Three extensions would carry this work furthest. **(1) Timestamped performances.** Archival All India Radio broadcast schedules, printed festival programmes (all-night events such as the Dover Lane Music Conference are ideal, since they span every prahar), and timestamped concert videos would supply real (raga, time, place) triples, enabling the falsifiable headline test: does the solar formalization predict programming better than the fixed clock? As a by-product they would give the first quantitative measurement of samay observance in modern practice. **(2) Corpus scale.** The CompMusic Hindustani raga-recognition corpus (about 300 recordings, 30 ragas) removes the single-raga-window confound, supports the TDMS baseline comparison directly, and lifts K to a size reviewers of raga recognition expect. Carnatic corpora are not an appropriate extension, since Carnatic practice largely does not observe time prescriptions. **(3) Validated emotion.** The psychometric rating study specified alongside this paper would replace synthetic tokens with reliable rasa and valence-arousal labels, restoring the affective axis to the framework on an evidential footing.

Conclusion. We have given the Hindustani samay system an explicit solar formalization anchored where the tradition itself is solar, the twilight ragas, and shown by exact integration that the choice between this formalization and the fixed-clock convention of prior computational work affects more than one hour in seven of the median Indian listener's day. We have released the first source-cited, machine-readable raga–samay table with uniform treatment of anytime, light, seasonal, and imported ragas, and shown on an open corpus that the standard tonic-normalized pitch representation, computed from expert annotations, recovers raga identity far above chance under leave-one-artist-out evaluation with recording-level significance testing. We have tried to be equally clear about what the evidence does not show: recovering one's own formalization validates no practice, synthetic tokens are not emotions, annotation-derived features certify no end-to-end system, and a window class carried by a single raga tests nothing. Timestamped programming data and a listener rating study would cross that line; both are specified here and ready to run.

Data and code availability. Both notebooks (solar/divergence; Saraga pipeline), the 91-raga table, and all seeds are released at `https://github.com/sudiptai/solar-samay-raga`. Every

figure and number in this paper regenerates from a single run of each notebook.

A Identifiability of the formalization (simulation)

Scope. This appendix asks a design question: if a system computed the features of Section 3, could it recover the raga prescribed by the formalization? Queries are generated from the ψ-arcs and a synthetic 5-way rasa token, so the labels are a noisy function of the very variables some encodings receive. Results therefore measure recoverability of our own construction and the sensitivity of feature encodings; they say nothing about music, listeners, or performance practice.

Design. 20 *fixed*-status ragas from the unified table (their corrected windows), 13 cities, uniform day-of-year; each query samples a clock time whose ψ falls in the raga's arc at that place and date, and emits the raga's token with probability 0.8 (else uniform). 6,000 queries per run, 70/15/15 splits stratified by raga, a two-hidden-layer network (64-32, dropout 0.15, Adam, early stopping), and four encodings: raw clock ($\sin / \cos t$), clock+token, solar ($\sin / \cos \psi$, prahar one-hot, lat/lon), solar+token. Paired contrasts are repeated over five data-generation seeds, each redrawing queries, split, and initialization, and are reported as mean $\pm$ Monte-Carlo standard error with sign consistency; with $n = 5$ repetitions, rank tests have no resolution and are omitted.

Closed-form references and results. Because the generative process is known, the right references are closed-form rules, not learned baselines. The Bayes-optimal rule attains 0.611 top-1 (0.972 top-3) on the reference run; a lookup rule (ragas whose arc contains ψ, filtered by token, ties broken at random) attains 0.634 on one draw, but over 300 tie-breaking redraws it scores 0.619 ± 0.012 (range 0.589–0.656), a distribution that contains the Bayes rule: the two are tie-breaking variants of one optimal family, and any single-test-set ordering between them is noise. The network's encodings score 0.254 (clock), 0.586 (clock+token), 0.307 (solar), and 0.619 top-1 / 0.957 top-3 (solar+token) on the reference run; the full model thus matches the closed-form optimum, and in simulation the learning component contributes nothing beyond recovering the known rule, as expected. Paired over five seeds, the token adds $+31.0 \pm 0.4$ points top-1 (a measurement of the simulator's noise parameter, reported only as such), and the solar encoding adds $+3.9 \pm 0.7$ with the token and $+4.2 \pm 0.4$ without; every contrast is positive in all five seeds and exceeds twice its Monte-Carlo standard error. Per-city gains are descriptive only: an apparent correlation between gain and distance from the IST meridian in an earlier version vanished under seed-averaging ($r = 0.00$, $p = 0.99$; mean gain +4.4 points for cities more than 6° from the meridian versus +4.0 within). The gap between top-1 (0.619) and top-3 (0.957) reflects that several ragas legitimately share an arc and token; the task is many-to-one by design, which is why Section 6 frames deployment as ranked selection under a soft prior.

Table 2: Identifiability simulation, reference run (20 ragas; the "token" is synthetic, not an emotion label), with closed-form references. Paired contrasts over five seeds appear in the text.

Encoding	Top-1	Macro-F1	Top-3
clock	0.254	0.180	0.748
clock + token	0.586	0.540	0.938
solar	0.307	0.224	0.849
solar + token	0.619	0.584	0.957
rule lookup (expected over tie-breaks)	0.619 ± 0.012	–	–
Bayes-optimal (no learning)	0.611	–	0.972

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